EconomicBridge — Technical Operator's Handbook

A plain-English guide to how every component works, so you can explain any part of the platform with confidence. Read Part 1 for the big picture, Part 2 for the data inputs (satellites + APIs), Part 3 for the seven modules, Part 4 for how it all runs, and Part 5 for quick answers to likely questions.

PART 1 — THE BIG PICTURE

One sentence: EconomicBridge collects live data from satellites and global data providers, processes it with AI and statistics, stores it per-tenant, and serves it as maps, alerts and reports — then pushes warnings to communities by SMS.

The data journey (left to right):

```
SATELLITES & APIs   ->  INGESTION   ->  DATABASE   ->  API   ->  DASHBOARD
(Copernicus, NASA,     (downloads,     (PostgreSQL  (serves   (maps, charts,
 World Bank, UNICEF,    processes,      schema-per-  clean     alerts) + SMS
 WorldPop, HDX, N2YO)   scores, tags)   tenant)      JSON)     to communities
```

The five services (each a separate program, all on AWS):

Service	Job
frontend	The dashboard people see (Next.js web app + Mapbox maps)
api	Serves data to the dashboard, handles login, tenants, reports
ingestion	Pulls satellite/economic data on a schedule, processes it
ml	Runs the AI models (crop disease, conflict, anomalies)
notifications	Sends SMS alerts (multilingual), gated by data-protection rules

The golden rule — honest provenance: every figure is tagged with where it came from. Live satellite/measured data is labelled distinctly from *modelled* baseline estimates. The dashboard never dresses up an estimate as a measurement. This is a core trust feature, and a strong thing to emphasise to any technical reviewer.

PART 2 — THE INPUTS: HOW EACH SATELLITE / API WORKS

For each source: **what it is**, **what it measures**, **how we use it**, **freshness**.

Copernicus Sentinel-1 — radar (SAR)

- **What:** A European Space Agency radar satellite. SAR = Synthetic Aperture Radar.
- **How radar works (the key point):** It sends its own microwave pulses down and measures what bounces back. Because it makes its own "light," it works **day or night and sees straight through cloud** — unlike a normal camera. Smooth water reflects the pulse away (looks dark/low return); rough ground and vegetation scatter it back (bright). That contrast is how we detect **standing flood water** and land-surface change.
- **We use it for:** ShockGuard flood detection, and Farmland Protection (land-surface change / encroachment signal).
- **Freshness:** ~weekly per area (revisit cycle); we pull it via Copernicus's Statistical API, which returns numbers (averages over an area) rather than raw image files — fast and light.

Copernicus Sentinel-2 — optical / vegetation (NDVI)

- **What:** ESA optical satellite (a very sophisticated camera across many colour bands, including near-infrared).
- **What it measures — NDVI:** Healthy plants reflect a lot of near-infrared light and absorb red light. NDVI (Normalised Difference Vegetation Index) = (NIR – Red) / (NIR + Red). High NDVI = lush, healthy vegetation; falling NDVI = stress, disease, drought, or harvest.
- **We use it for:** CropGuard's satellite vegetation monitoring (crop health week by week, no photo needed) and ShockGuard drought signal.
- **Freshness:** ~weekly; also via the Statistical API (we get the NDVI value over each area, not raw pixels).

NASA FIRMS — active fire (thermal)

- **What:** NASA's Fire Information for Resource Management System.
- **What it measures:** Satellites (MODIS, VIIRS) detect **thermal anomalies** — spots much hotter than their surroundings = active fires/bush burning.
- **We use it for:** Farmland Protection (fire near farmland is a conflict/loss signal) and as a hazard layer.
- **Freshness: Daily**, near-real-time (within hours of the satellite pass).

VIIRS Black Marble — night-lights (poverty proxy)

- **What:** NASA's night-time lights product (collection VNP46A2), accessed with a NASA Earthdata token.
- **What it measures:** How brightly the Earth glows at night. **Lit areas = economic activity and electrification; dark-but-populated areas = poverty.**
- **We use it for:** Economic Visibility — scoring which settlements are poor and underserved (people present, little light).
- **Freshness:** Weekly in our scheduler (night-lights move slowly for this purpose).

WorldPop — population grids

- **What:** Open gridded population estimates (~100 m squares), from the University of Southampton.
- **What it measures:** How many people live in each square of land.
- **We use it for:** Economic Visibility — turning "this settlement is dark" into "and ~X thousand people live there," so relief can be targeted by real need.
- **Freshness:** Weekly sampling. (We serve the raster from our own secure S3 copy because WorldPop's public server stopped supporting partial reads — a good example of engineering around an upstream problem.)

N2YO — satellite pass tracking

- **What:** A live satellite-tracking service.
- **What it does for us:** Tells us **when a satellite will next fly over a given area**, so the ingestion service can request fresh imagery around that pass ("pass-driven refresh") instead of guessing.
- **Freshness:** Checked every 15 minutes.

World Bank Indicators API — economics

- **What:** The World Bank's open data API (free, CC BY 4.0 — commercial use OK).
- **What it measures:** National economic indicators — GNI per capita, employment.
- **We use it for:** Mobility Compass — income and cost-of-living, anchored in USD and converted to each country's local currency.
- **Freshness:** Monthly.

UNICEF GIGA — schools & connectivity

- **What:** UNICEF's global school-mapping initiative.
- **What it measures:** Locations of schools and their internet connectivity.
- **We use it for:** SkillsBridge — education access and connectivity targeting.
- **Freshness:** Monthly. (Real per-area school counts are live; fine-grained connectivity is partly modelled — labelled honestly.)

HDX HAPI — humanitarian / aid presence

- **What:** The UN OCHA Humanitarian Data Exchange API (keyless, open).
- **What it measures:** "Who does what where" — which aid agencies operate in which areas (operational presence).
- **We use it for:** Aid Coordination — showing coverage and gaps.
- **Freshness:** Monthly. (Honest note: its operational-presence data covers north-east Nigeria; elsewhere our coverage view leans on operator CSV uploads.)

The AI models (run by the ml service)

- **CropGuard ResNet-50:** A 50-layer convolutional neural network trained to recognise **12 crop-disease classes** from a leaf photo. We trained it to **87.2% validation accuracy**. It returns the top likely diseases with a

confidence score, and a **Grad-CAM heatmap** showing *where on the leaf* it looked — so a human can sanity-check it. Low-confidence results are flagged for human review rather than asserted.
- **Conflict predictor (Random Forest):** Combines signals (fire, land change, proximity, history) to estimate the probability of farmer-herder / encroachment conflict in the next **24–72 hours**. Above a confidence threshold it raises an alert; below it, it logs for review.
- **Anomaly detectors (statistics):** For floods and drought we compare the recent satellite reading to a baseline and compute a **z-score** (how many standard deviations from normal). A big jump = an anomaly worth flagging. Simple, robust, and explainable.

PART 3 — THE OUTPUTS: HOW EACH MODULE WORKS

For each module: **purpose · inputs · how it works · what you see · provenance.**

1. Economic Visibility (Poverty Mapping)

- **Purpose:** Find poor, underserved settlements that traditional census misses.
- **Inputs:** VIIRS night-lights + WorldPop population (+ DHS-style validation).
- **How:** Dark-but-populated settlements score as higher-poverty; population gives the scale of need.
- **You see:** A map of settlements coloured by poverty intensity, a vulnerability ranking, and counts (villages identified, population, households unreached).
- **Provenance:** Settlement layer is a modelled baseline (`seed`); population and night-lights enrich it live (`worldpop_cog_v1`, `viirs_v2`) → badge shows the blend.

2. Aid Coordination

- **Purpose:** Stop aid agencies duplicating effort and leaving gaps.
- **Inputs:** HDX HAPI operational presence + operator CSV uploads (WFP/UNHCR).
- **How:** Maps which agencies work where, against need, to surface gaps.
- **You see:** Coverage by area, agency lists, gap highlights.
- **Provenance:** `hapi_v1` where HDX covers; CSV-sourced elsewhere.

3. Farmland Protection

- **Purpose:** Warn of herder/cattle encroachment and farmer-herder conflict **before** it happens.
- **Inputs:** two live engines — (a) a **fire-cluster conflict pipeline** (NASA FIRMS heat clusters → Random-Forest conflict model), and (b) the **encroachment detector** that fuses **Sentinel-2 NDVI loss** (grazing/clearing) + **Sentinel-1 SAR change** (land-surface disturbance, tracks, trampling) + **FIRMS fire**.
- **How — encroachment detector:** we cannot see individual cattle at 10 m, so we detect their *signatures*. Per tenant ROI we compare recent satellite readings to a baseline; a single notable signal raises a **medium watch**, and corroborating signals (loss + radar + fire) escalate to **high/critical**. Confidence is corroboration-weighted, so a lone wet-season radar change (often just soil moisture) never reads as critical. Runs daily for every tenant, year-round — independent of fire season.
- **You see:** pulsing alert halos, a conflict-risk timeline, each alert labelled with an LGA + the real trigger ("vegetation loss", "radar land-surface change", "N fires"), flagged for human review.
- **Provenance:** live encroachment alerts (model_name=`encroachment_detector_v1`) vs seed baseline are labelled — LIVE when present, MONITORING when the watch is on but nothing's active. These are ROI-level *risk indicators*, not confirmed incidents; higher-resolution / per-LGA data (NASRDA NCRS) would localize them from ROI to field scale.

4. CropGuard

- **Purpose:** Protect harvests from disease, and watch crop health from space.
- **Inputs:** (a) a leaf photo → ResNet-50; (b) Sentinel-2 NDVI; (c) market prices; (d) yield signals.
- **How:** Two complementary capabilities — **point diagnosis** (photograph one sick leaf, AI names the disease in seconds with a heatmap) and **continuous satellite vegetation monitoring** (NDVI over the whole area, no photo needed, flagging stress/anomalies early).
- **You see:** The TRAINED model badge, a diagnosis with confidence + Grad-CAM, an NDVI vegetation panel, crop prices, and a yield outlook.

- **Provenance:** Model badge reads TRAINED (real model serving everywhere); NDVI panel defaults to live Sentinel-2.

5. ShockGuard

- **Purpose:** Detect floods and drought early.
- **Inputs:** Sentinel-1 SAR (flood) + Sentinel-2 NDVI (drought) + z-score detectors.
- **How:** Flood = sudden rise in smooth-water radar signature; drought = NDVI falling well below the seasonal baseline. Both expressed as anomalies with confidence.
- **You see:** Flood/drought events on the map with severity (critical / watch).
- **Provenance:** Runs on live Copernicus data; modelled fallback labelled if a tenant lacks acquisitions.

6. Mobility Compass

- **Purpose:** Economic picture — income, cost of living, displacement capacity.
- **Inputs:** World Bank GNI/employment, USD-anchored, converted per local currency.
- **How:** National indicators disaggregated to per-LGA estimates; dual-currency (e.g. USD + Naira) so figures are locally legible.
- **You see:** Income and cost-of-living indicators across LGAs.
- **Provenance:** `worldbank_v1` (real) over the `seed` baseline.

7. SkillsBridge

- **Purpose:** Target education and connectivity investment.
- **Inputs:** UNICEF GIGA school locations + World Bank ICT.
- **How:** Per-area school density and access mapped to find under-served areas.
- **You see:** School access / connectivity indicators per LGA.
- **Provenance:** `giga_v1` (real school counts) over baseline.

PART 4 — HOW IT ALL RUNS (OPERATIONS)

The ingestion scheduler (9 self-running jobs)

The ingestion service runs jobs automatically on a clock (no one has to press a button), and each can also be fired on demand from the Admin panel:

Job	Cadence
Pass-driven imagery refresh (N2YO + Sentinel)	every 15 min
NASA FIRMS fire ingest	daily 06:00 UTC
Conflict prediction sweep	daily 06:30 UTC
Sentinel-1 SAR + Sentinel-2 NDVI observations	weekly (Sat)
WorldPop population sweep	weekly (Sun)
VIIRS night-lights + poverty ingest	weekly (Mon)
World Bank income (Mobility)	monthly (1st)
HDX aid coordination	monthly (1st)
UNICEF GIGA schools (Skills)	monthly (1st)

The database — schema-per-tenant isolation

Each tenant (state/agency) has its **own schema** inside PostgreSQL. A request for Kebbi data physically cannot read Benue's. Regional partners (e.g. NEMA, ECOWAS) get cross-tenant read access by role. This is the core of the security story.

The cloud — AWS

Five containers run on **AWS ECS Fargate** behind one **load balancer** that routes by path (`/api`, `/ingestion`, `/ml`, `/notifications`, and the dashboard at `/`). Data in **RDS PostgreSQL**, caching in **ElastiCache Redis**, model weights and the WorldPop raster in **S3**, all secrets in **AWS Secrets Manager** (never in code). Everything is defined in **Terraform** and deployed via **GitHub Actions CI/CD**.

Security & compliance

- Secrets only in Secrets Manager; nothing sensitive in the code repo.
- JWT login (short-lived access + refresh tokens), per-IP rate limiting.
- An **INSERT-only audit log** records data changes.
- PII (e.g. subscriber phone numbers) sits behind a signed Data-Processing-Agreement gate; aligned with the Nigeria Data Protection Act.

The last mile — SMS

Warnings reach communities as plain **SMS in local languages** (English, French, Portuguese live; Hausa/Yoruba/Igbo in preparation) through partner agencies — **no smartphone, no app, no literacy barrier**. Farmers don't register; numbers come via partner agencies acting as data controllers.

PART 5 — QUICK ANSWERS (likely questions)

- **"Is this really live or a mock-up?"** → Live in production on AWS; 700+ LGAs; here's the URL. Live data is labelled distinctly from modelled baselines.
- **"Where does your satellite data come from?"** → Copernicus Sentinel-1 (radar) and Sentinel-2 (NDVI), NASA FIRMS and VIIRS, plus WorldPop, World Bank, UNICEF GIGA, HDX. We'd love to add **NigeriaSat**.
- **"How accurate is the AI?"** → The crop model is a ResNet-50 at 87.2% validation, with a heatmap and human-review gating; conflict uses a Random Forest with confidence thresholds; floods/drought use statistical anomaly detection.
- **"How does radar see through clouds?"** → It's active microwave — it makes its own signal and reads the echo, so cloud and darkness don't block it. Water looks dark to radar, which is how we map floods.
- **"How do you detect herder/cattle encroachment?"** → We can't see individual cattle at 10 m, so we detect their *signatures*: Sentinel-2 vegetation loss (grazing/clearing) + Sentinel-1 radar land-surface disturbance + FIRMS fire, fused per ROI into a year-round risk watch. A lone signal is a medium watch; corroborating signals escalate to high/critical in the dry/conflict season. Higher-resolution NigeriaSat / NCRS data would localize the watch from ROI to the actual field, and confirm cattle vs. soil moisture.
- **"How is each state's data protected?"** → Schema-per-tenant isolation in the database; a state can never read another's; regional bodies get cross-state view by role.
- **"What do you want from us (NASRDA)?"** → Integrate NigeriaSat data, run a joint demonstration, and explore a data-sharing MOU — your satellites, visibly protecting Nigerian farmers.
- **"What's the business model?"** → Government/agency subscriptions (free pilot → annual licence); farmer SMS stays free, funded by institutional subscriptions.

PART 6 — DEEP DIVE FOR A GIS / REMOTE-SENSING AUDIENCE

Read this before meeting a GIS professional. These are the details they will probe, in their own language.

Coordinate systems

All data is handled in **WGS84 geographic coordinates (EPSG:4326, lat/lon)**. The web map renders in **Web Mercator (EPSG:3857)** via Mapbox GL, with deck.gl drawing the data layers on top. Tenant areas are defined by a **region-of-interest (ROI) bounding box**: `[min_lon, min_lat, max_lon, max_lat]`.

Administrative boundaries & spatial joins

- Admin-2 units (LGAs / districts) come from **geoBoundaries** (open data).
- Each settlement / event point is assigned to its LGA by a **point-in-polygon spatial join**; per-LGA **centroids** drive map placement and the per-tenant lists (Kebbi 21, Benue 23, Kaduna 23, Niger 25, Plateau 17, Zamfara 14, Nasarawa 13, FCT 6 area councils, Ghana 260, Senegal 45).

Raster handling (WorldPop)

- WorldPop population is a **GeoTIFF / Cloud-Optimized GeoTIFF (COG)**, ~100 m.
- We **point-sample** it at each settlement's coordinates with **rasterio**, handling **nodata** cells.

- We serve the raster from **our own S3 mirror** because the upstream server stopped honouring HTTP range requests — which breaks COG windowed reads. A GIS pro will appreciate that detail: we restored proper range-read behaviour rather than downloading whole 100–500 MB rasters per query.

Zonal statistics (Sentinel via Copernicus Statistical API)

Instead of downloading raw scenes, we call Copernicus's **Statistical API**, which computes **server-side zonal statistics** (mean / min / max / percentiles) over each tenant's ROI per time bucket. So Sentinel-1 returns backscatter stats and Sentinel-2 returns NDVI stats, as **time series** ready for anomaly detection — efficient, and no gigabytes of imagery moved.

Sensor specifics

Sensor	Type	Resolution	Measure / bands	Revisit
Sentinel-1	C-band SAR	~10 m	VV/VH backscatter	~6–12 days
Sentinel-2	Optical	10–20 m	NDVI = (B8 – B4)/(B8 + B4)	~5 days
VIIRS Black Marble	Night optical	~500 m	radiance (VNP46A2)	daily (we use weekly)
WorldPop	Modelled raster	~100 m	persons / pixel	annual
NASA FIRMS	Thermal	375 m–1 km	active-fire detections	daily (NRT)

NDVI math

NDVI = (NIR – Red) / (NIR + Red); on Sentinel-2 that is **(B8 – B4)/(B8 + B4)**. Range –1 to +1; dense healthy crops sit ~0.6–0.9. We compare a recent mean against a baseline mean and flag the **z-score** deviation.

SAR for flood

Open water is a **specular reflector** — it bounces the radar pulse away from the sensor, so it returns **very low backscatter (dark)**. A sudden drop in VV/VH backscatter over normally-bright land indicates **standing water** → mapped flood extent, cloud-independent and day or night.

Anomaly detection (the maths, plainly)

Per ROI we hold a baseline distribution; each new acquisition is scored **z = (recent_mean – baseline_mean) / baseline_std**. Beyond a threshold → anomaly, with a confidence band. Deliberately transparent — no black box for hazard calls.

Visualization stack

Mapbox GL JS basemap (satellite / dark) + **deck.gl** GPU layers for points and heatmaps. Pulse animations are decoupled from the data layers so re-rendering stays smooth.

Honest GIS caveats (knowing these earns credibility)

- We currently place per-LGA data at **centroids** for display; full-polygon **choropleth** is on the roadmap.
- The Statistical API gives **ROI-aggregate** values, not per-pixel rasters — excellent for trends, not for sub-LGA pixel mapping.
- Some layers (fine connectivity, the settlement baseline) are **modelled** and labelled as such.

Where NASRDA / NigeriaSat fits (the GIS framing for the meeting)

NigeriaSat-2 (~2.5 m panchromatic, ~5 m multispectral) and the **NCRS** archives would give us **higher-resolution, locally-owned scenes** to: map at **sub-LGA / field scale**, **validate** our 10 m Sentinel-derived indicators against finer imagery, and move from ROI-aggregates toward **full-polygon analysis**. In GIS terms: NigeriaSat complements our medium-resolution, high-revisit Sentinel pipeline with high-resolution detail — a genuine capability gain, not a duplication. *Lead with this when the director asks what you'd do with their data.*
